

Towards biharmonic maps of degree 1 from $\mathbb{T}^2$ to $\mathbb{S}^2$

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1. Harmonic maps

Harmonic maps play a central role in differential geometry; They are defined as the critical points of the energy functional

$$E(\phi) = \frac{1}{2} \int_M |d\phi|^2 v_g$$

for smooth maps ϕ from (M, g) into (N, h) , closed Riemannian manifolds.

- ϕ is harmonic if and only if its tension field vanishes:

$$\tau(\phi) = \text{tr}_g \nabla d\phi = 0.$$

- Generalizes geodesics (for 1-dimensional M)
- Generalizes harmonic functions. ($N = \mathbb{R}$)

In 1976, Eells & Wood [2] showed that there exists no harmonic map from $\mathbb{T}^2$ to $\mathbb{S}^2$ (whatever the metrics chosen) in the homotopy classes of Brouwer degrees ± 1 .

An alternative argument was later given by Schoen & Yau [4], based on the Bochner’s formula for the harmonic map equation in complex form and the Gauss-Bonnet theorem.

Theorem 1. *Suppose M, N are closed. If $|\bar{\partial}u|$ is not identically zero, then*

$$\sum_{\substack{p \in M \\ |\bar{\partial}u|(p)=0}} m_p = \deg(u)(2g_N - 2) + (2g_M - 2),$$

where m_p is the order of $|\bar{\partial}u|$ at p .

Corollary 1.1. *Suppose $M, \mathbb{S}^2$ are equipped with arbitrary metrics, and $N = \mathbb{S}^2$. If $u : M \rightarrow \mathbb{S}^2$ is harmonic and $\deg(u) > g_M - 1$, then u is holomorphic.*

Corollary 1.2. *There is no harmonic map of degree 1 from $\mathbb{T}^2$ to $\mathbb{S}^2$.*

Proof. • Suppose u is harmonic with $\deg(u) = 1$

- $\deg(u) > g_{\mathbb{T}^2} - 1 = 0 \xrightarrow{\text{Cor 1.1.}} u$ is holomorphic

- Then $J(u) = |\partial u|^2 - |\bar{\partial}u|^2 \geq 0$

- Since $\deg(u) = 1$ then $\#(u^{-1}(q)) = 1$, for any regular value $q \in \mathbb{S}^2$

- u is holomorphic, then u is a branched covering

- Since $\deg(u) = 1$, there are no branches and hence $J(u) > 0$ everywhere

- Therefore u is a diffeomorphism $\Rightarrow \Leftarrow$

□

2. Biharmonic maps

In 1983, J. Eells and L. Lemaire extended [3] the notion of harmonic maps to biharmonic maps, which

are defined as the critical points of the bienergy functional

$$E_2(\phi) = \frac{1}{2} \int_M |\tau(\phi)|^2 v_g. \quad (1)$$

- ϕ is biharmonic iff its bitension field vanishes:

$$\tau_2(\phi) = \bar{\Delta}(\tau(\phi)) - \text{tr}_g R^N(\tau(\phi), d\phi) d\phi = 0.$$

- every harmonic map is biharmonic
- non-harmonic biharmonic maps are called proper biharmonic
- they arise naturally in elasticity, submanifold theory, and generalizations of minimal surfaces.

The case studied by Eells & Wood changes once we consider biharmonicity. This question remains to be answered [1].

3. Our purpose

To study the biharmonic maps construction from $\mathbb{T}^2$ to $\mathbb{S}^2$ of degree 1.

Idea: Represent $\mathbb{T}^2$ as a quotient space of a square with opposite edges identified. Construct a map

$$[-1, 1]^2 \rightarrow \mathbb{D}^2 \rightarrow \mathbb{R}^2 \rightarrow \mathbb{S}^2$$

Requirements:

- Well defined on $\mathbb{T}^2$
- Continuous
- Onto $\mathbb{S}^2$
- Boundary $\rightarrow$ North pole
- Composition of homeomorphisms
- Orientation-preserving ($\deg = \pm 1$)
- Non-constant on open sets
- C^∞ on $\mathbb{T}^2$

1. Square to Unit Disk:

Let $(x_1, x_2) \in [-1, 1]^2$. Then define

$$T(x_1, x_2) := \left(x_1 \sqrt{1 - \frac{x_2^2}{2}}, x_2 \sqrt{1 - \frac{x_1^2}{2}} \right)$$

- Homeomorphism onto its image
- C^∞ local diffeomorphism on $(-1, 1)^2$
- T sends the square boundary to the unit circle
- Preserves orientation in the interior

2. Disk to Plane:

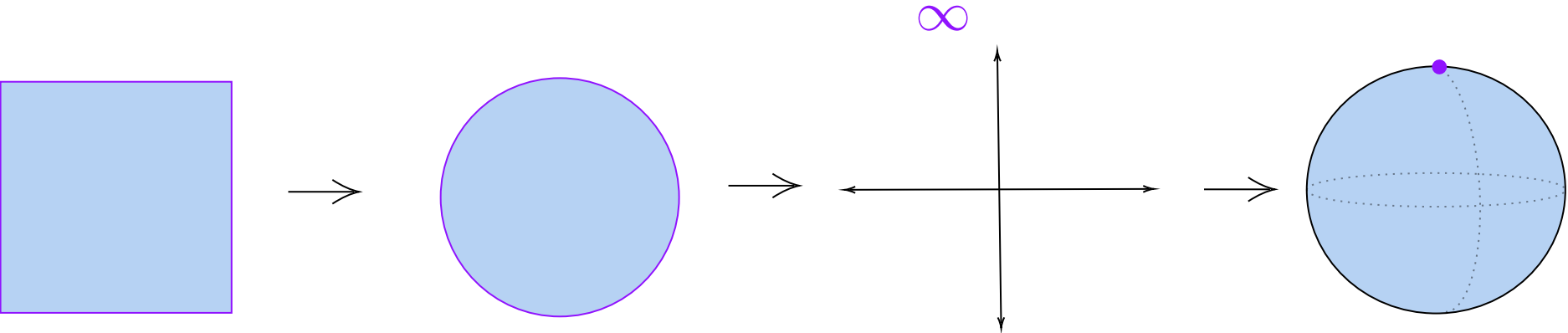
$$g(z) = \begin{cases} \tan\left(\frac{\pi}{2} \|z\|\right) \frac{z}{\|z\|}, & z \neq 0 \\ 0, & z = 0 \end{cases}$$

- Diffeomorphism from the unit disk to $\mathbb{R}^2$
- $\|z\| \rightarrow 1 \Rightarrow g(z) \rightarrow \infty$ in direction of z
- Orientation-preserving

3. Plane to Sphere (inverse stereographic proj.):

$$\theta(w_1, w_2) = \left(\frac{-1 + w_1^2 + w_2^2}{1 + w_1^2 + w_2^2}, \frac{2w_1}{1 + w_1^2 + w_2^2}, \frac{2w_2}{1 + w_1^2 + w_2^2} \right)$$

- Diffeomorphism from $\mathbb{S}^2 \setminus \{\text{north pole}\}$ to $\mathbb{R}^2$
- $\lim_{\|(x,y)\| \rightarrow \infty} \theta(x, y) = (1, 0, 0)$.
- Preserves orientation.



4. Composition:

Let $x = (x_1, x_2) \in [-1, 1]^2$. Set $A := \sqrt{x_1^2 + x_2^2 - x_1^2 x_2^2}$.

Then

$$F(x) = \begin{cases} (-\cos(\pi A), s(A)v_1(x), s(A)v_2(x)), & \|x\|_\infty \neq 1 \\ (1, 0, 0), & \text{otherwise,} \end{cases}$$

with $s(A) = \frac{2 \sin(\pi A)}{A}$, and $v_i(x) = x_i \sqrt{1 - \frac{x_i^2}{2}}$, $i \neq j$.

- Fulfils all (almost) the requirements, but partial derivatives does not match at boundaries.

To regularise their derivatives at the boundary, we redefine the map as

$$F_{sm}(x) = (-\cos(\pi \hat{A}), s(\hat{A})v_1(x), s(\hat{A})v_2(x))$$

with $\hat{A}(x_1, x_2) = 1 - \chi(\rho(x_1, x_2))(1 - A(x_1, x_2))$, where $\rho(x_1, x_2) = (1 - x_2^2)(1 - x_1^2)$ is a smooth boundary defining function, and $\chi(t)$ a standard flat bump given by

$$\chi(t) = \frac{b(t)}{b(t) + b(1 - t)}, \quad t \in [0, 1]$$

where

$$b(t) = \begin{cases} e^{-\frac{1}{t}}, & t > 0 \\ 0, & t \leq 0. \end{cases}$$

4. Forthcoming Research

Finally, there will be able to delve deeper into the biharmonicity issue, calculating its tension field, bitension field, analysing how to define a variation of this map F_{sm} that satisfies (1).

- Use symmetries and reduction methods.

References

[1] J. Eells and J. H. Sampson. Variational theory in fibre bundles. In *Proceedings of the United States–Japan seminar in differential geometry, Kyoto, Japan, 22–33*, 1965.

[2] J. Eells and J. C. Wood. Restrictions on harmonic maps of surfaces. *Topology* 15, 263–266, 1976.

[3] L. Eells, J.; Lemaire. Selected topics in harmonic maps. In *Proceedings of the CBMS Regional Conference Series in Mathematics, Providence, RI, USA*, 1983.

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